Bina.az Housing Price Prediction

From-scratch decision trees and Pegasos SVM

MLE-AI-201 Final Project V3

PegaSold

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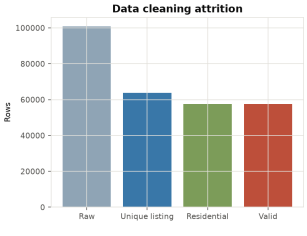
Project question — Speaker 1: data and task

- Bina.az asking-price scrape: 100,775 raw rows; October–November 2024 records.
- Task A predicts $\ln(\mathrm{price})$; errors are reported in log space and AZN.
- Task B predicts standard/premium; the train-only median cutoff is 208,000 AZN.
- Decision tree and Pegasos training are NumPy implementations; sklearn supplies comparison baselines.

Cleaning and leakage control — Speaker 1

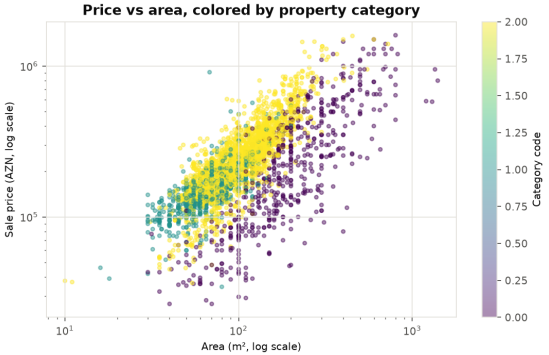
- Normalize advert URLs to numeric IDs; keep each advert's latest scrape.
- Keep Baku's three residential categories; fixed bounds check price, floor area, AZN/m^2, rooms and floors.
- Train-derived log-price MAD fence: 26,130–1,671,651 AZN; removed 246 from train only.
- Splits after train filtering: 40,033/8,632/8,632; validation/test retain all rows.
- Exclude target-derived values, IDs, free text, owner/agency fields and raw address.
- Imputation, scaling and location vocabulary





Exploration and feature design — Speaker 1

- Raw EDA measures missingness, repeat scrapes, leakage candidates, categories, geography and price tails.
- Parse Azerbaijani numeric strings; represent yes/no flags as 0/1 and invalid coordinates as missing.
- Use log area/land, location/type indicators and engineered distance/floor/area ratios.
- EDA describes this 2024 scrape; it is not causal evidence or today's market estimate.

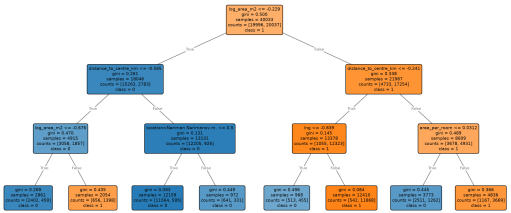


Split, targets, and validation protocol — Speaker 1

- Seed 42: 40,033 train, 8,632 validation, 8,632 test rows.
- Repeated advert IDs are kept in one partition; split allocation balances price bins.
- Premium threshold is independently learned from training prices only: 208,000 AZN.
- Select hyperparameters on validation, refit on train, and evaluate test after selection.
- Five-fold CV uses all 40,033 training rows; shared preprocessing precedes folds, so CV is not fully nested.

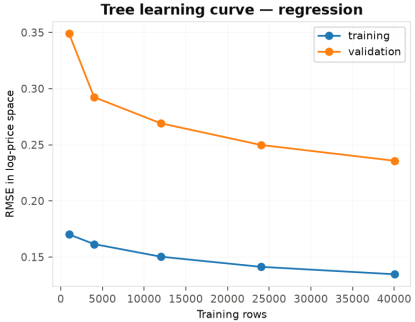
Decision tree from scratch — Speaker 2

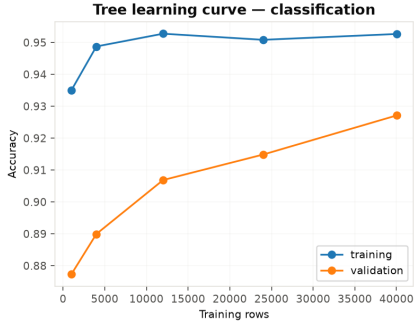
- Sort numeric values and evaluate boundaries between distinct adjacent values.
- Cumulative class counts compute Gini/entropy gain; target sums and squared sums compute MSE reduction.
- Recurse on the best legal split; leaves predict majority class or mean target.
- Depth, minimum leaf and split constraints control model growth.
- Importance sums sample-weighted impurity decreases; it does not establish causation.



Tree results and generalization — Speaker 2

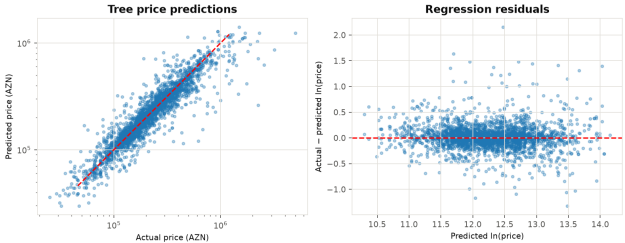
Metric	Score	Metric	Score
Task A log RMSE	0.239	Task B accuracy	0.922
Task A AZN RMSE	147,356	Task B F1 / AUC	0.923 / 0.958
Reference tree log RMSE	0.239	Reference accuracy	0.922





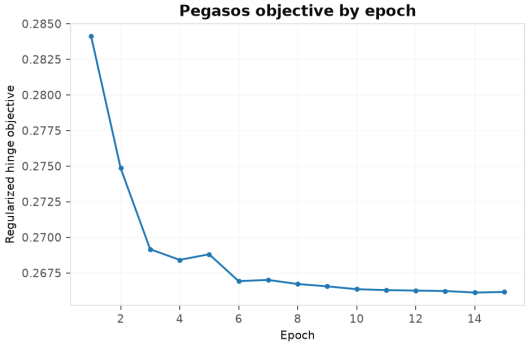
Interpreting the tree's AZN price error — Speaker 2

- The tree learns mean squared error on $\ln(\text{price})$, then exponentiates its output to AZN.
- Test MAE: 47,613 AZN (average); median absolute error: 16,431 AZN.
- Test RMSE: 147,356 AZN; maximum absolute error: 3,760,714 AZN.
- The maximum test error is 3,760,714 AZN: one 5,000,000 AZN home was estimated at 1,239,286 AZN.
- Validation/test retain prices outside the fence; luxury-tail errors remain large and visible.
- This is a historical asking-price



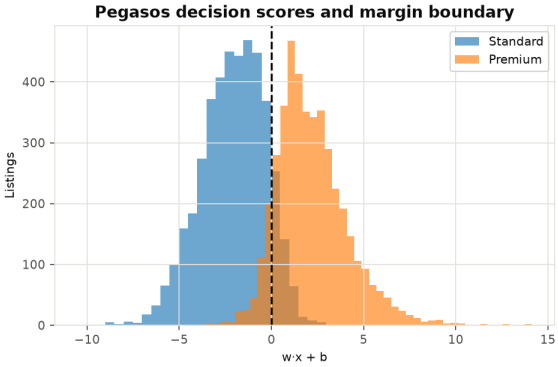
Linear Pegasos SVM — Speaker 3

- Minimize $\lambda\|w\|^2/2+$ mean hinge loss using stochastic subgradient updates.
- Standardize from training data; labels map to $-1, +1$ internally.
- Selected $\lambda = 0.01$; margin width $2/\|w\| = 0.9001$.
- Support vectors (signed margin ≤ 1): 11,587 training rows.
- Test: accuracy 0.902, F1 0.903, ROC-AUC 0.968.



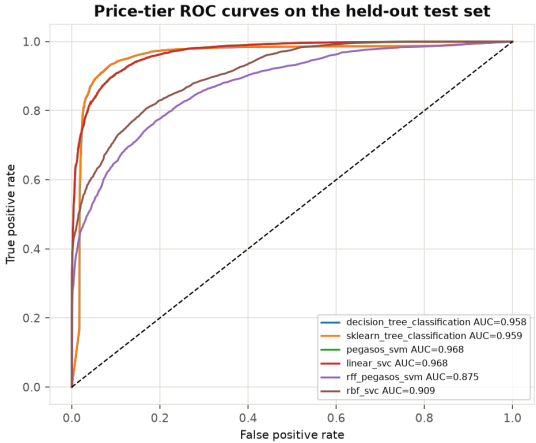
Nonlinear SVM and decision margin — Speaker 3

- Random Fourier Features use seeded cosine projections to approximate an RBF kernel; NumPy Pegasos trains the mapped features.
- 128 components; selected $\gamma = 0.01$; tuned on an 8,000-row training subset.
- RFF test: accuracy 0.790, F1 0.786, AUC 0.875.
- Linear model performed better on this dataset/run; nonlinear mapping did not guarantee a gain.



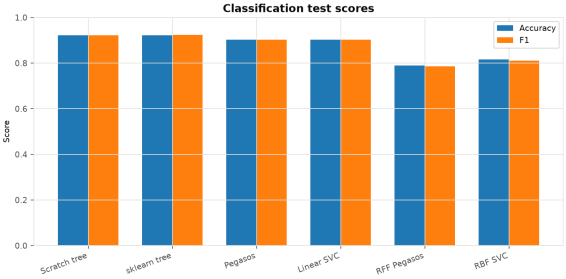
Fair comparison and uncertainty — Speaker 4

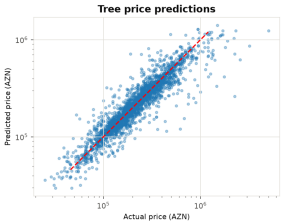
- Tree references use matching criteria/depth/leaf settings on the same input matrix.
- LinearSVC uses hinge loss, train-only scaling pipeline and objective-matched $C = 1/(\lambda n)$.
- Five-fold CV is on a seeded train sample; report means and standard deviations.
- Test metrics are held out once; test results do not drive model selection.

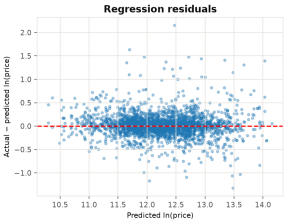


Held-out classification comparison — Speaker 4

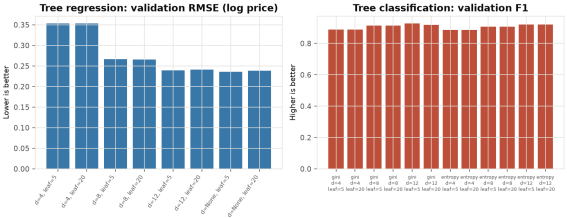
Model	Accuracy	F1	AUC	Fit sec
Scratch tree	0.922	0.923	0.958	1.76
sklearn tree	0.922	0.923	0.959	0.24
Pegasos	0.902	0.903	0.968	2.81
LinearSVC	0.902	0.903	0.968	0.48
RFF Pegasos	0.790	0.786	0.875	0.82



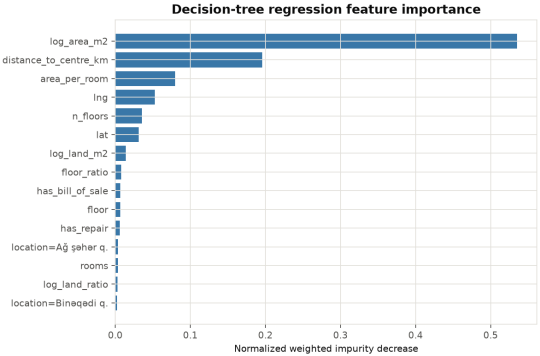




Validation sweeps and model interpretation — Speaker 4



Validation parameter sweeps.



Regression tree importance.

Validation chooses settings; importance measures split utility and is not causal evidence.

Conclusions and limitations — All

- Median price-tier classification is easier than exact advertised-price prediction; AZN error exposes expensive-property misses.
- Asking prices are not verified transaction prices; one 2024 scrape is not today's market.
- Fixed cleaning bounds were chosen after source EDA; ambiguous cross-ID duplicate candidates may remain.
- Sparse locations and the non-nested CV preprocessing can limit generalization claims.
- Reproduce with `python -m src.run_all`; full results and caveats are in `reports/final_report.md`.

Reproducibility and four-part handoff — All

1. Speaker 1: raw EDA, cleaning, targets, splits and leakage checks.
2. Speaker 2: NumPy decision tree, criteria, tuning and tree results.
3. Speaker 3: Pegasos, RFF mapping, margin and SVM findings.
4. Speaker 4: baselines, cross-validation, metrics, error analysis and reproducibility.

Run and verify

```
python -m src.run_all
pytest -q
```

The automated test summary and outputs are in `reports/` and `data/processed/`.

AI-assisted coding suggestions, debugging, corrections and review were used; the team checked the final code and results.